Exciting Developments in the Analysis of Big Data

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Disclosures

- NIH Grants
  - NS065257
  - GM113852
- Statistical Editor
  - *Anesthesiology*
  - *Headache*

Exciting Developments

- Specification
  - Variable selection
  - Machine Learning
  - Ensemble Models

- Estimation
  - Variational inference
  - Hamiltonian Monte Carlo (NUTS)

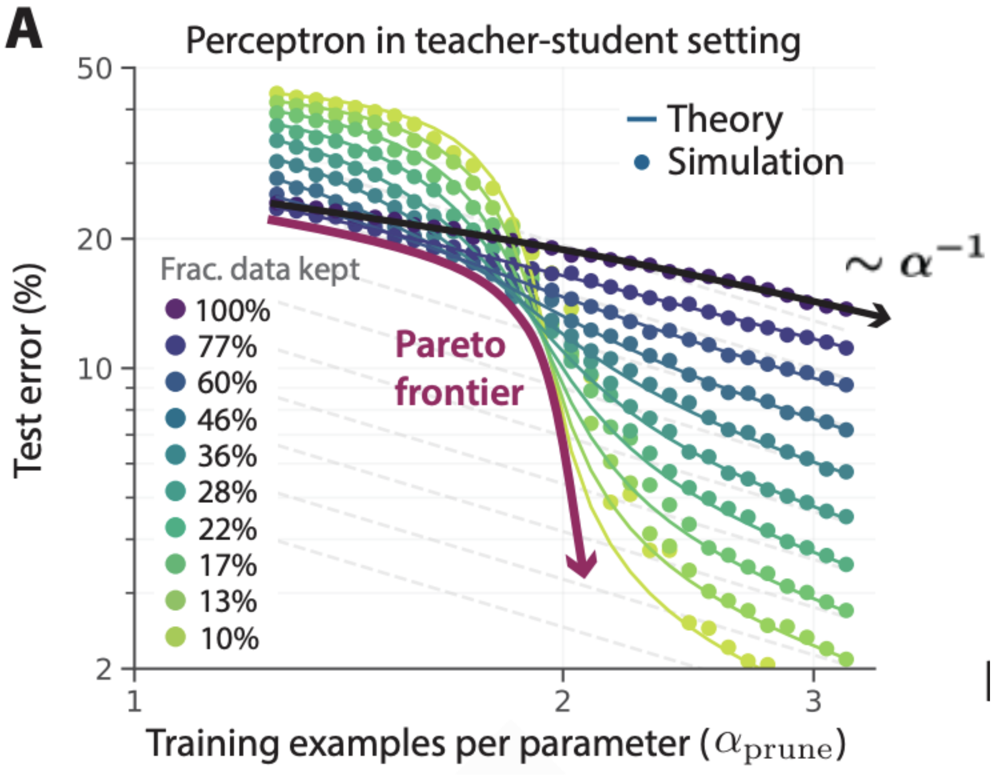
- Interpretation
  - DAG
  - Sensitivity analysis (e.g., vibrational-analysis)

Probabilistic Programming

https://livebook.manning.com/book/practical-probabilistic-programming/chapter-1/

Beyond neural scaling laws: beating power law scaling via data pruning

Reducing model error
From 3% to 2% requires
an order of magnitude more
data (i.e., power law)



By curating training data, rather than random sampling, this
ratio can be reduced to a mere exponential

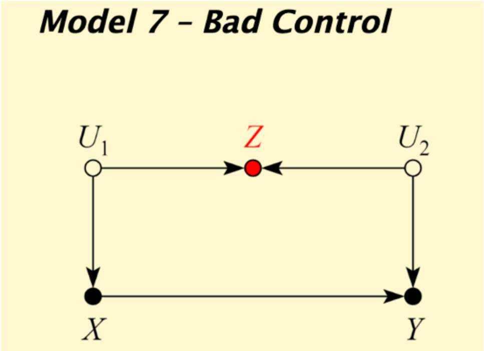
https://arxiv.org/pdf/2206.14486.pdf

Exciting Developments in ‘The Big 3’

- Confounding
- Missing Data
- Measurement Error

Confounding & Bias

- Curated list of common sources of bias and confounding (with examples!)
  - https://catalogofbias.org/
- 'Good' and 'Bad' control
  - http://causality.cs.ucla.edu/blog/index.php/category/bad-control/
- 'Table 2 fallacy'
  - https://academic.oup.com/aje/article/177/4/292/147738

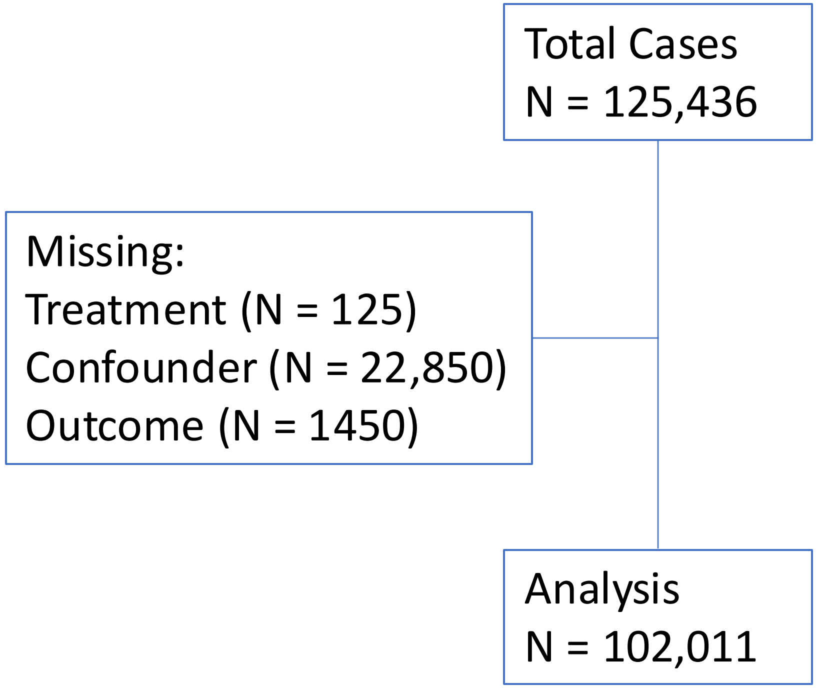


VanderWeele (2019). Principles of Confounder Selection. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6447501/

Missing Data

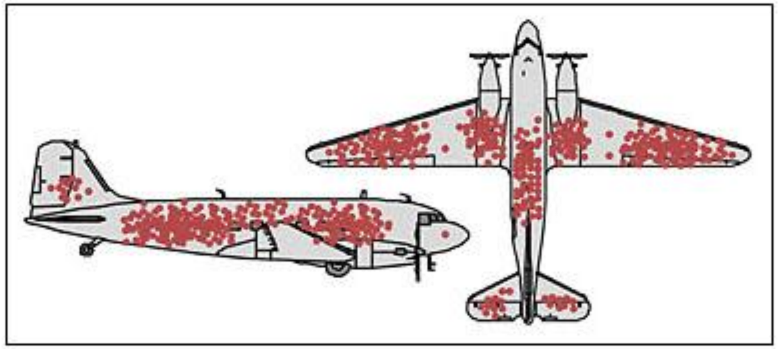
- Missing data is very common in large data bases
  - Core element of the data generating process
  - Random element of data entry or elicitation

- Most common strategy is to ignore missing data
  - Complete case analysis
  - Eekhout et al. (2012).



Abraham Wald’s WWII Bomber Analysis

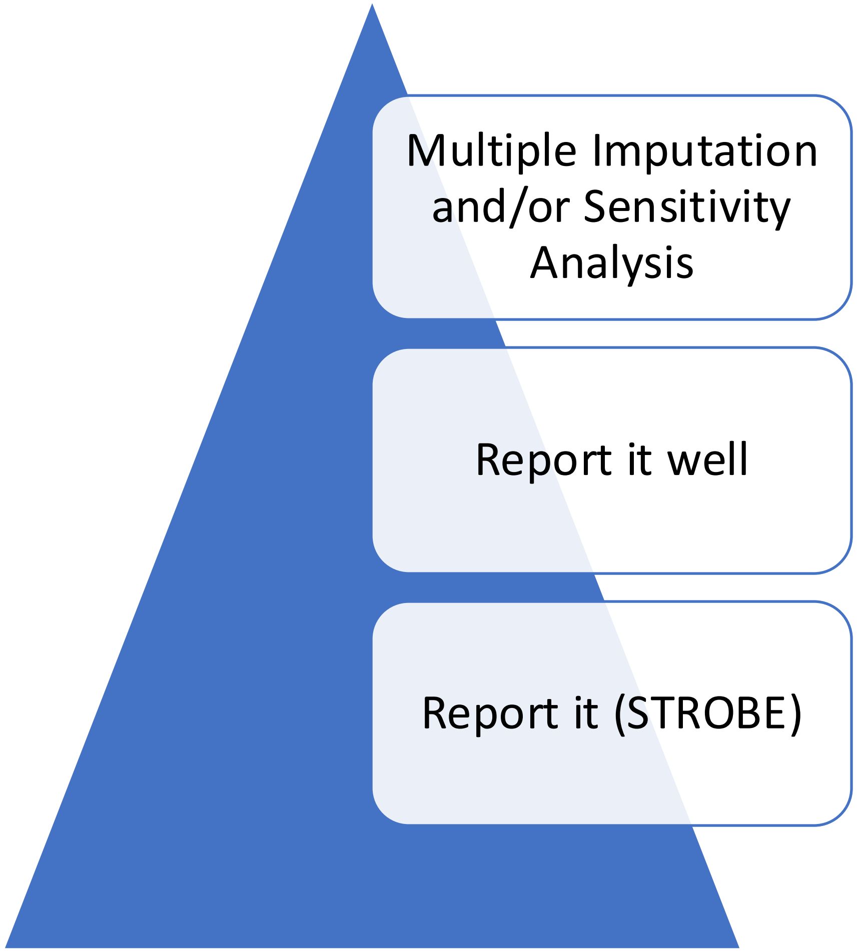
- Where to armor the bombers?
- Study of “hit analysis”
- Armor where bullets are NOT!



The Endeavor Blog (J. Cook)

What to do about missing data?

- Report it (STROBE)
  - “Explain how missing data were addressed”
- Report it well
  - Predictors of missing-ness
- Imputation
  - Likelihood-based approaches
  - Weighted estimation
  - Multiple imputation
- Sensitivity analyses
  - What if?
  - Best case, worst case



Measurement Error

- Mismeasurements and misclassifications of all kinds
  - Mistaken entries
  - Inaccurate recordings
  - Imperfectly reliable measurements

- Observed = True + Error

Measurement error is often neglected in medical literature: a systematic review

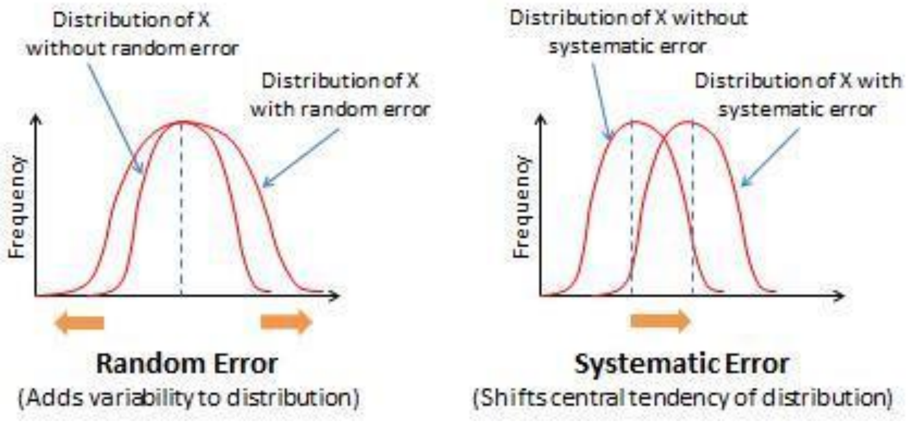
- Original research published in 2016 in high-impact medical and epidemiology journals
  - Main exposure or confounder

- 247/565 (44%) directly addressed measurement error
  - 70% ONLY in the Discussion section

Brakenhoff TB, et al. Measurement error is often neglected in medical literature: a systematic review. Journal of clinical epidemiology. 2018 Jun 1;98:89-97.

Types of Measurement Error

- Classical error
  - ~ N(0,σ2)
- Systematic error
  - ~N(bias, σ2)
- Differential
  - Error dependent on outcome
- Berkson
  - True + ~N(0, constant)



Measurement Error Myths

- Measurement error can be compensated by large number of observations
  - Increased N causes estimates to approach the measurement error mechanism, not their true value
- The exposure effect is *underestimated* when variables are measured with error
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- Measurement error can be prevented but not mitigated in observational data analysis
- Certain types of observational research are unaffected by measurement error

Measurement Error Correction Methods

- Regression calibration (Rosner et al., 1989)
  - Error prone covariate replaced by expected true score
- Simulation-extrapolation (SIMEX; Cook et al., 1994)
  - Simulate dataset through adding MORE error to the covariates
  - Extrapolate predictions back to original situation
- Latent variable models
  - Replicate measures used to estimate ‘true’ value of a more reliable latent construct
- Bayesian approaches

Recommendations

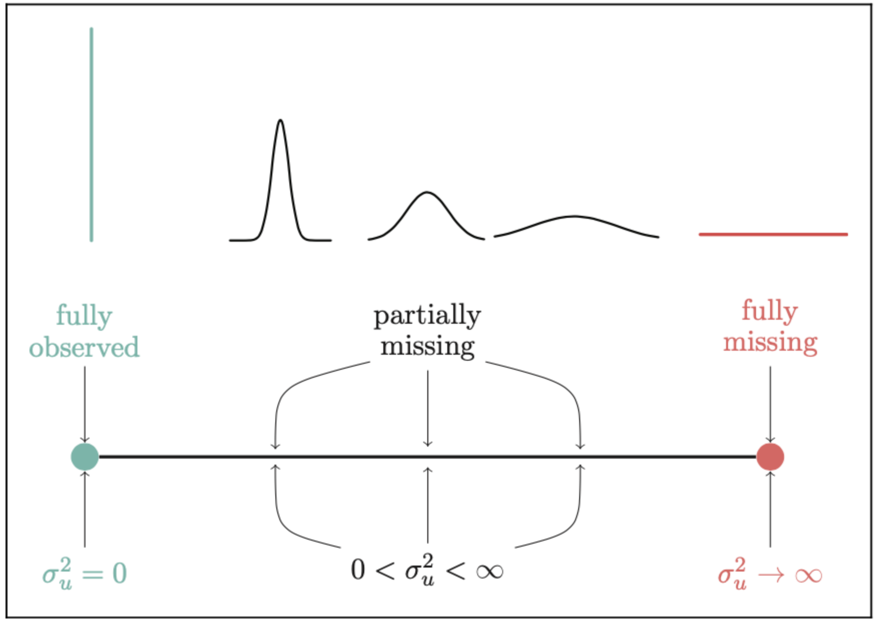
- Measurement error is nearly ubiquitous in observational data analysis
  - Let’s stop neglecting it
- Measurement error can have a counter-intuitive impact on observed associations
  - Must consider its structure AND degree
  - Be cautious when applying general claims about the direction of the error
- Strongly consider the use of formal strategies to mitigate error
  - Conduct validation efforts
  - Utilize formal statistical methods

van Smeden M, Lash TL, Groenwold RH, van Smeden M. Five myths about measurement error in epidemiologic research.

A Unified Approach to Measurement Error and Missing Data

- Multiple Imputation for Measurement Error (MIME)
  - Cole et al. (2006): https://academic.oup.com/ije/article/35/4/1074/686404

- Multiple Overimputation (MO)
  - Blackwell et al., (2017)



Keogh et al. (2019). Measurement Error as a Missing Data Problem:
https://arxiv.org/abs/1910.06443

Thank you!

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